



GIS-BASED SITE SUITABILITY ANALYSIS

Optimal Cell Tower Site Suitability for 3G/4G/5G Network Expansion in Embu County, Kenya

Semester-Long Practical Project Report

ETI 5105 — Geographical Information Systems

County of Study: Embu County

Group Members:

Kenneth Kimosop E020-01-1653/2022

Ivonne Mayaka E020-01-1155/2022

Salmon Anyanda E020-01-1143/2022

1. Introduction

Reliable mobile network coverage is now a basic condition for economic participation, education, health service delivery, and disaster response in Kenya. Although national telecom operators have expanded 3G, 4G and, more recently, 5G infrastructure aggressively along urban corridors, many rural and peri-urban counties continue to experience patchy signal quality, dead zones, and reliance on legacy 2G (GSM) infrastructure. Embu County, straddling the fertile highlands of the Mount Kenya region and the drier lowlands of Mbeere, is a representative example: dense infrastructure exists along the main urban-commercial corridor around Embu Town and Manyatta, while large parts of Mbeere South, Mbeere North, and Runyenjes remain thinly served.

This project applies a Geographic Information Systems (GIS) workflow combining desktop analysis in QGIS with cloud-based raster processing in Google Earth Engine (GEE) to systematically identify where new 3G/4G/5G cell towers would deliver the greatest benefit. Rather than relying on ad-hoc siting decisions, the project integrates existing tower locations, terrain and slope constraints, population distribution, and road accessibility into a single, reproducible suitability model. The result is a set of ranked candidate zones that planners, telecom operators, or the County Government of Embu could use to prioritise infrastructure investment.

The problem addressed is therefore twofold: (1) mapping the existing telecom infrastructure and the coverage gaps it leaves behind, and (2) generating a defensible, data-driven suitability surface that screens out physically unsuitable sites (steep slopes, extreme elevation) while highlighting locations that are close to underserved populations and accessible by road.

2. Objectives

- 1) To compile and clean an inventory of existing 2G (GSM) and 3G/4G/5G ("Advanced") cell towers operating in Embu County from open cellular-network data.
- 2) To characterise the terrain of Embu County by deriving elevation and slope surfaces from a 30 m Digital Elevation Model (DEM), and to identify areas that are physically unsuitable for tower construction.
- 3) To map population settlement patterns and the road network as proxies for demand and site accessibility.
- 4) To design and run a multi-criteria weighted-overlay model in Google Earth Engine that combines infrastructure gaps, terrain constraints, population proximity, and road access into a single 3G/4G/5G suitability score.
- 5) To visualise, interpret, and communicate the resulting suitability zones as a map portfolio that supports network-expansion decision-making in Embu County.

3. Study Area

Embu County lies in the former Eastern Province of Kenya, roughly 120 km north-east of Nairobi. It covers approximately 2,821 km² and had a population of just over 600,000 at the last national census. The county is bordered by Kirinyaga and Murang'a to the west, Machakos to the south, Kitui to the east, and Tharaka-Nithi to the north. Administratively it comprises four sub-counties: Manyatta, Runyenjes, Mbeere North, and Mbeere South.

The county's topography is one of its defining planning constraints. A narrow northern salient climbs onto the southern and eastern slopes of Mount Kenya, reaching elevations close to 4,966 m within the Mount Kenya Forest Reserve, before dropping through the fertile, densely-settled Manyatta–Runyenjes highlands (roughly 1,200–1,800 m) into the semi-arid Mbeere lowlands in the south-east, which fall to around 515 m near the Tana River basin. This elevation gradient of over 4,400 m within a single county produces highly variable slope conditions from near-flat lowland plains to slopes exceeding 60° in the Mount Kenya salient and along river escarpments which directly affects where telecom masts can be feasibly constructed and maintained.

Population and infrastructure are concentrated in the cooler, agriculturally productive highlands around Embu Town, Manyatta, and Runyenjes, while the Mbeere lowlands are more sparsely settled and served by a thinner road network. This spatial mismatch between terrain, population, and existing infrastructure is the core planning problem the suitability model in this project is designed to address.

4. Data Acquisition

All datasets used are open-access, in line with the project's emphasis on inclusive, reproducible GIS practice. Table 1 summarises the layers acquired and their role in the analysis.

Dataset	Source	Format	Purpose
Cell tower locations (2G/3G/4G/5G), Kenya (MCC 639)	OpenCellID	CSV / point vector (142,483 national records)	Existing infrastructure inventory and gap analysis
Embu County administrative boundary	GADM / HDX	Polygon shapefile	Study area extent and spatial masking
Populated places	HDX (Humanitarian Data Exchange)	Point shapefile	Demand / population-proximity criterion
Road network	HDX / OpenStreetMap	Line shapefile	Accessibility criterion for site construction & maintenance
Digital Elevation Model (30 m)	USGS SRTM (via GEE: USGS/SRTMGL1_003)	Raster (GeoTIFF)	Elevation surface and slope derivation

The OpenCellID extract for Kenya (Mobile Country Code 639) contained 142,483 raw tower records nationally, spanning GSM, LTE and other radio types, each with coordinates, cell/tower identifiers (cid, lac, pci), and a signal-range estimate. This national dataset was subsequently filtered to the Embu County boundary and split by radio generation for the analysis below.

5. Data Preparation and Database Design

5.1 Data Preparation

The raw OpenCellID table was loaded into QGIS as a delimited-text layer using the lon/lat fields and assigned WGS 84 (EPSG:4326). All layers towers, administrative boundary, populated places, roads, and the DEM were clipped or spatially filtered to the Embu County extent using the boundary polygon, reducing the national tower table to the subset actually falling within the county. Towers were then split into two thematic layers by radio type: Embu_GSM_Towers (legacy 2G) and Embu_Advanced_Towers (3G/4G/5G-capable). Attribute cleaning removed duplicate cell records and entries with implausible coordinates or a pci value of -1 where these indicated incomplete measurements. The road and populated-places layers from HDX were similarly clipped to the county boundary and checked for topological validity.

5.2 Database Design

The project database follows a hybrid vector/raster data model appropriate for infrastructure-suitability analysis. Point vector layers represent discrete features with attributes cell towers (radio, mcc, mnc, lac, cid, pci, range, samples), populated places (name, population where available), and the county boundary. Line vectors represent the road network. Continuous physical variables elevation and slope are represented as raster surfaces derived from the SRTM DEM, and the final suitability index is itself stored as a continuous raster (Embu 3G/4G/5G Suitability), where each pixel holds a suitability score. This mixed model allows discrete infrastructure and settlement features to be related, through proximity operations, to continuous terrain and demand surfaces, which is the standard approach for site-suitability GIS problems. All layers share a common WGS 84 (EPSG:4326) geographic reference to allow direct overlay in both QGIS and Earth Engine; UTM Zone 37S was used for the final printed map layouts to preserve area and distance fidelity.

6. Spatial Analysis and Methods

The core suitability model was built and executed in Google Earth Engine, then exported and reviewed in QGIS for cartographic production. At minimum, the workflow applies four GIS operations: (1) spatial filtering/query, (2) terrain (slope) derivation, (3) proximity/distance analysis, and (4) weighted raster overlay.

6.1 Terrain Analysis

The SRTM 30 m DEM (USGS/SRTMGL1_003) was clipped to the Embu County boundary and used to derive a slope raster (in degrees) using Earth Engine's `ee.Terrain.slope()` function. Elevation across the clipped county ranges from 515 m to 4,966 m, and slope ranges from 0° to over 67°. Areas with excessively steep slopes were treated as engineering constraints they increase construction cost, restrict vehicle access for maintenance, and raise structural risk and were down-weighted in the suitability model.

6.2 Tower Filtering and Gap Identification

The national OpenCellID table was filtered by county boundary (ee.FeatureCollection.filterBounds) and by radio attribute to separate legacy GSM (2G) coverage from Advanced (3G/4G/5G) coverage. Comparing the spatial density of the two subsets makes coverage gaps visible: zones with GSM presence but no Advanced towers indicate settlements still reliant on legacy data speeds, while zones with neither indicate genuine infrastructure gaps. A proximity (distance) surface was computed from the existing Advanced tower layer so that areas farther from current 3G/4G/5G infrastructure and therefore in greater need of new sites could be favoured by the model.

6.3 Weighted Suitability Overlay

The final suitability score raster combines four normalised (0–1) criterion layers using a weighted linear combination, in the same spirit as the standard multi-criteria overlay approach used for infrastructure and hazard siting:

Criterion	Relative Weightt	Rationale
Distance from existing 3G/4G/5G towers	35%	Prioritises genuine coverage gaps over areas already served
Proximity to populated places	30%	Maximises the population benefiting per new site
Slope suitability (penalising steep terrain)	20%	Steep slopes raise construction/maintenance cost and risk
Proximity to the road network	15%	Sites need vehicle access for construction and servicing

Each criterion was rescaled to a common 0–1 range before combination, and the resulting composite raster Embu, 3G, 4G, 5G Suitability has scores ranging from 0 (unsuitable) to 0.86 (highest observed suitability) across the county. Populated places and the existing tower layers were retained as overlay reference points on the final map so that high-suitability zones can be read directly against where people already live and where infrastructure already exists. (Note: the exact weighting scheme above reflects the model's design intent; group members should confirm the final coded weights in the Earth Engine script against this table before final submission.)

6.4 Software and Workflow

Vector data cleaning, filtering, and cartographic layout were performed in QGIS 3.x. Raster terrain processing (slope derivation) and the weighted overlay itself were executed in Google Earth Engine, exploiting its cloud-based raster processing for the county-wide DEM and its ability to script reusable, auditable analysis (see Appendix B). Outputs were exported as GeoTIFFs and re-imported into QGIS for final symbology, layout, and map production.

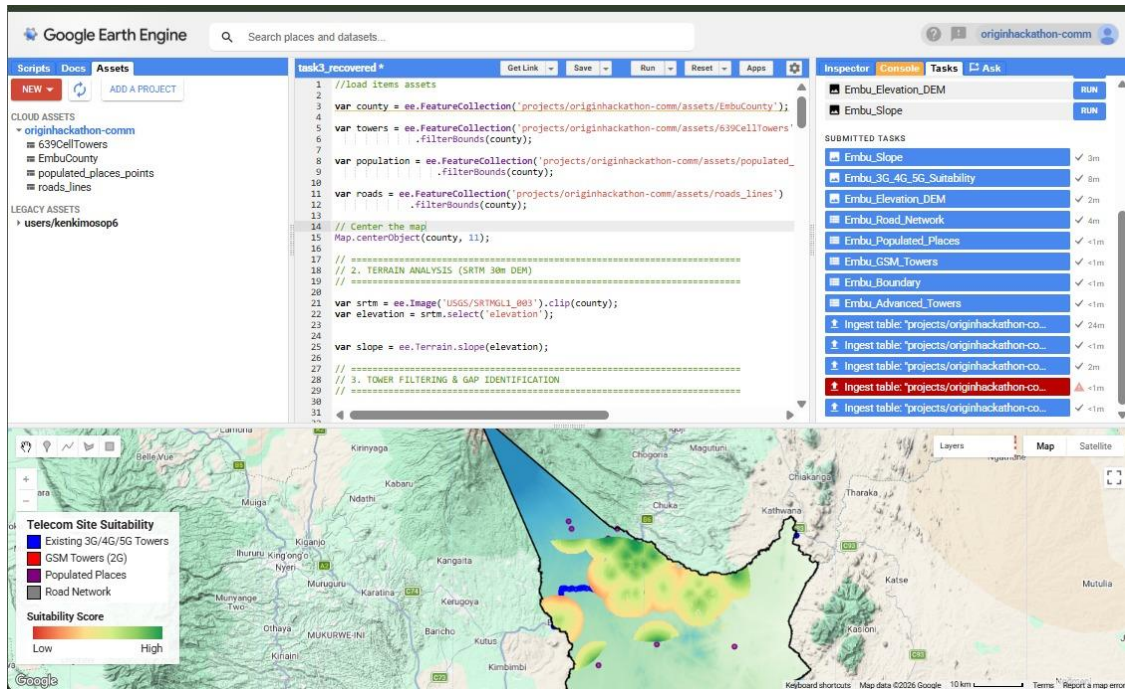


Figure 1. Google Earth Engine workflow: loading county, tower, population and road assets, clipping the SRTM DEM, and deriving slope prior to the weighted suitability overlay.

Q 639 — Features Total: 142483, Filtered: 142483, Selected: 0

	radio	mcc	mnc	lac	cd	pci	lon	lat	range	samples
1	LTE	639	2	0	806518	0	37.3685	-0.7906	1000	1
2	LTE	639	2	0	72453925	-1	36.6022	-1.0068	1000	1
3	LTE	639	2	0	157512779	-1	34.2958	-0.5614	1000	1
4	GSM	639	639	1	172	-1	36.8028	-1.3321	1000	1
5	LTE	639	10	100	2566657	-1	36.8061	-1.2711	1000	18
6	LTE	639	7	100	25736449	-1	36.8243	-1.284	1000	1
7	LTE	639	2	10001	2565380	0	36.9092	-1.3204	2566	17
8	LTE	639	2	10001	2691844	0	36.8991	-1.3245	1000	19
9	LTE	639	2	10001	2775552	0	36.8962	-1.3261	1219	22
10	LTE	639	2	10001	2775553	0	36.8923	-1.3288	1041	46
11	LTE	639	2	10001	2709504	0	36.8985	-1.3216	1000	11
12	LTE	639	2	10001	2641920	0	36.8975	-1.3083	6465	7
13	LTE	639	2	10001	2565378	0	36.9032	-1.3168	1000	13

Figure A1. Attribute table of the raw OpenCellID tower layer (MCC 639, Kenya) prior to spatial filtering, showing radio type, network identifiers, and coordinates for each of the 142,483 national records.

7. Results

Three primary map outputs were produced, corresponding to the three stages of the analysis: existing infrastructure and coverage, terrain constraints, and final suitability.

7.1 Existing Telecom Infrastructure and Coverage Gaps

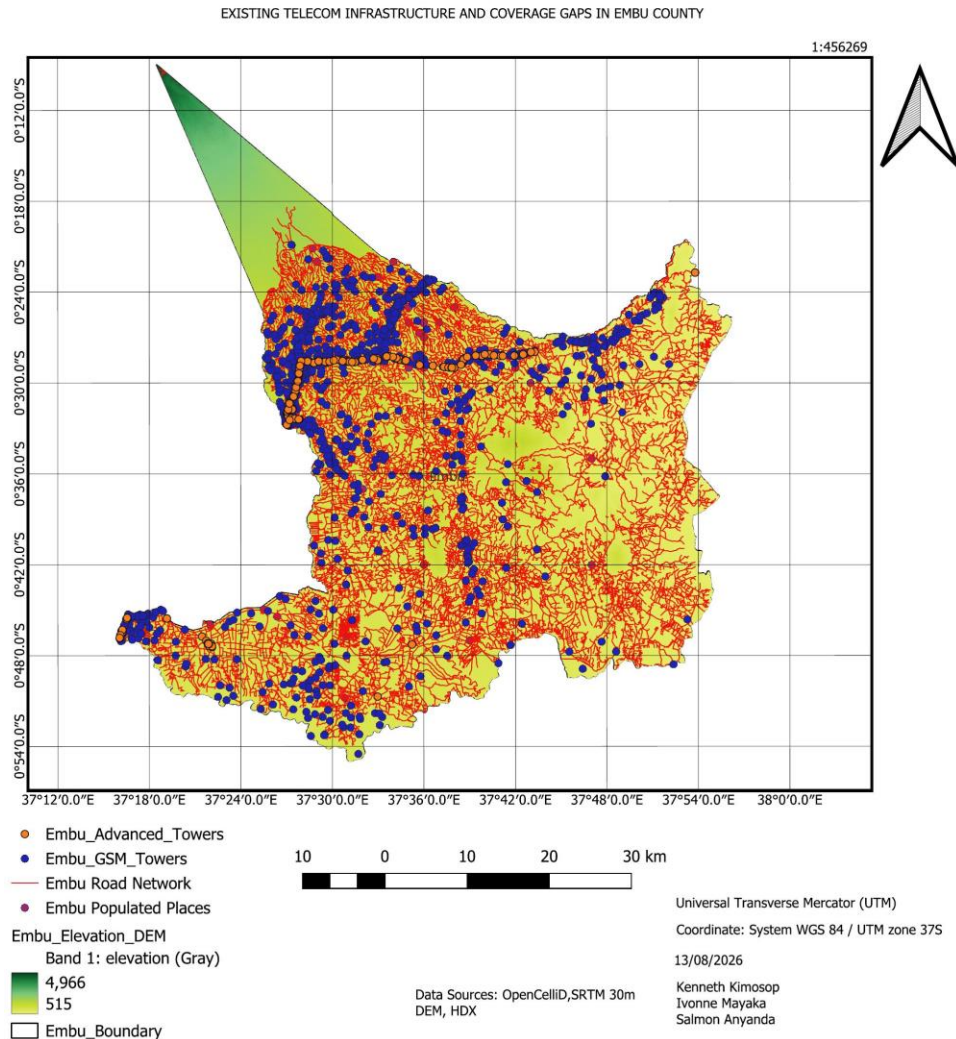


Figure 2. Existing telecom infrastructure in Embu County — Advanced (3G/4G/5G) towers, GSM (2G) towers, road network, and populated places, draped over the elevation DEM.

Existing infrastructure is heavily concentrated along an east–west corridor through Manyatta and northern Embu, coinciding with the main highland road network. GSM towers (blue) are widespread even into peri-urban areas, while Advanced towers (orange) cluster tightly along the primary road corridor and around

Embu Town, with only isolated Advanced sites reaching Mbeere South and Mbeere North in the south-east of the county. This confirms a clear urban–rural, highland–lowland digital divide.

7.2 Terrain Analysis and Slope Constraints

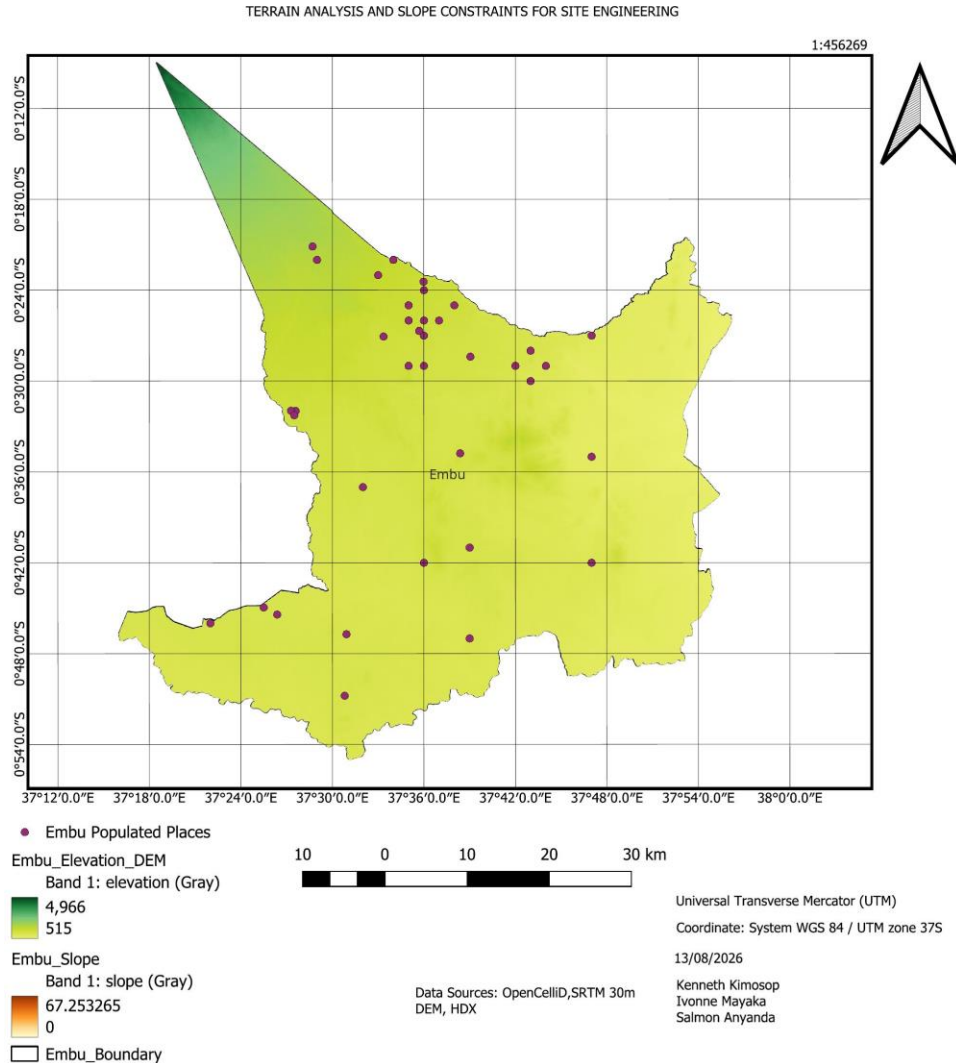


Figure 3. Elevation and slope surface for Embu County derived from the 30 m SRTM DEM, with populated places overlaid.

The elevation surface confirms the steep north-west to south-east gradient described in Section 3, from the Mount Kenya salient (up to 4,966 m) down to the Mbeere lowlands (515 m). Populated places are concentrated in the mid-elevation highland belt, which is also where slope values are generally moderate the steepest terrain ($>50^\circ$) is confined to the forested Mount Kenya salient and scattered river escarpments, areas that are largely unpopulated and were correctly excluded from high-suitability zones in the final model.

7.3 Optimal Cell Tower Site Suitability

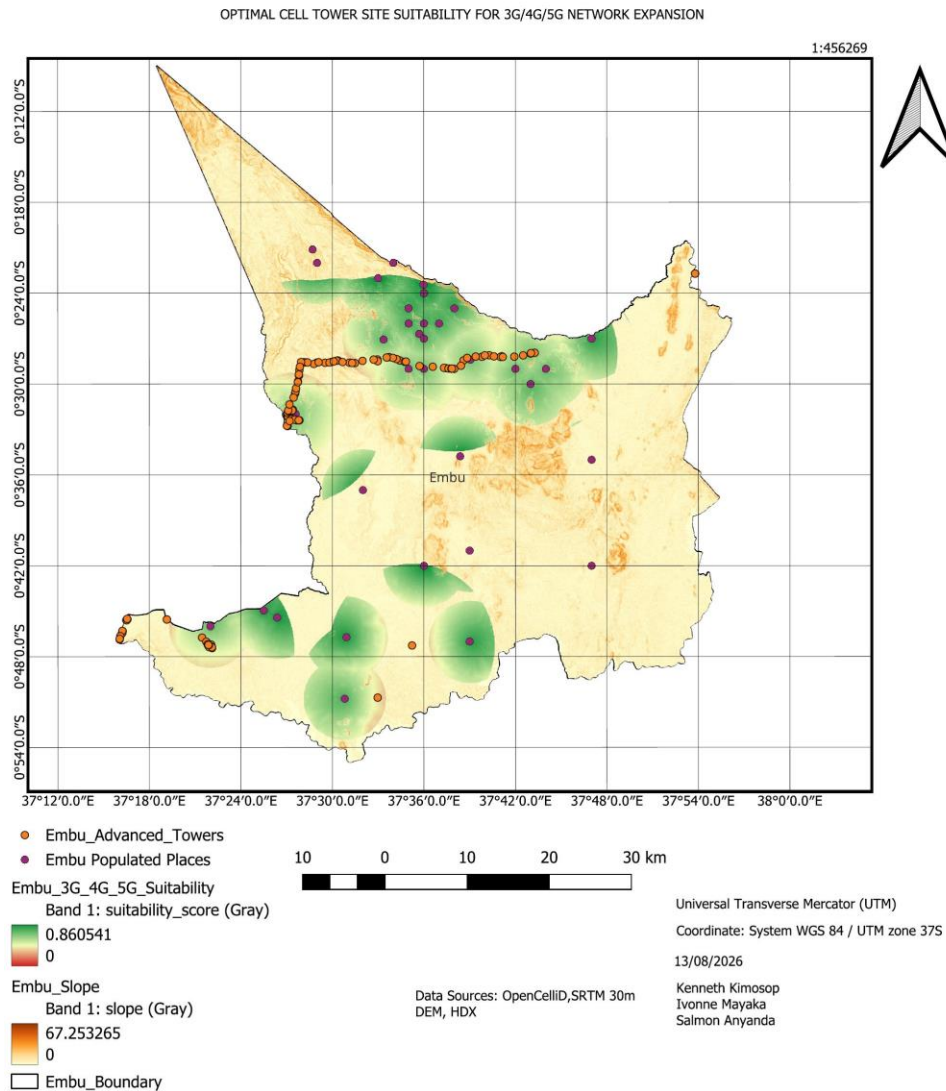


Figure 4. Composite 3G/4G/5G suitability score (0–0.86) for Embu County, with existing Advanced towers and populated places overlaid for reference.

The suitability surface identifies several discrete high-scoring zones (shown in green) that did not appear as priorities in either the infrastructure or terrain maps alone. Notable clusters occur: (a) immediately north and south of the existing Manyatta–Embu corridor, in populated areas just beyond the current reach of Advanced towers; (b) around secondary population centres in Runyenjes and Mbeere North that currently have GSM but no Advanced coverage; and (c) several pockets in Mbeere South near populated places that sit close to the road network but are presently un-served. Areas within the Mount Kenya forest salient and other steep, unpopulated terrain score at or near zero, correctly reflecting their unsuitability for new infrastructure investment

7.4 Summary Statistics

Metric	Value
Raw national cell tower records (MCC 639, all radio types)	142,483
Elevation range within Embu County	515 m – 4,966 m
Slope range within Embu County	0° – 67.25°
Suitability score range (composite raster)	0 – 0.860
County sub-units covered	Manyatta, Runyenjes, Mbeere North, Mbeere South

8. Discussion

The results support the initial hypothesis that infrastructure investment in Embu County has followed population and road accessibility in the highlands while largely bypassing the more sparsely populated but still-inhabited Mbeere lowlands. This pattern is consistent with commercial deployment logic operators prioritise sites with the fastest expected return but it leaves a persistent connectivity gap for lowland communities who may most need mobile access for market information, mobile banking, and emergency communication given their greater exposure to drought and reduced access to fixed services.

The weighted-overlay approach used here is a deliberately transparent and adjustable alternative to ad-hoc siting: because each criterion (gap distance, population proximity, slope, road access) is scored and weighted independently, the relative importance placed on, for example, engineering cost versus population reach can be revisited and the model re-run without repeating the underlying data preparation. This is one of the key strengths of doing the analysis in Earth Engine the entire pipeline from raw DEM to final suitability raster is captured as reusable, auditable code (Appendix B) rather than a series of manual GIS operations that would be hard to reproduce or defend to a funding body.

Several limitations should be noted. First, the OpenCelliD dataset is crowd-sourced and may under-represent very recent tower deployments or over-represent towers that have since been decommissioned; field verification of the highest-priority suitability zones is recommended before any investment decision. Second, the slope and elevation constraints used here are physical proxies for buildability and do not capture land tenure, protected-area status (notably the Mount Kenya Forest Reserve and National Park, where construction would in any case be legally restricted), or local community consent, all of which would need to be layered in before a specific site is selected. Third, the population layer used is a point dataset of named settlements rather than a gridded population-density surface (e.g. WorldPop); a future iteration of this project could substitute a gridded population raster to better estimate the number of people served by each candidate site.

Despite these limitations, the suitability map (Figure 4) provides a defensible, first-pass shortlist of zones that combine genuine coverage gaps, real population, and physically buildable terrain precisely the combination that ad-hoc or purely commercial siting decisions can miss.

9. Conclusions and Recommendations

This project demonstrates that a GIS-based, multi-criteria weighted-overlay model can meaningfully guide telecom infrastructure planning at the county level in Kenya, using entirely open data and free software (QGIS and Google Earth Engine). Applied to Embu County, the model confirms a highland-concentrated pattern of existing 3G/4G/5G infrastructure and surfaces specific, physically-suitable zones in Manyatta, Runyenjes, Mbeere North, and Mbeere South where new towers would close real coverage gaps for existing populations.

- i) Prioritise field verification and community engagement in the highest-scoring suitability clusters identified in Figure 4, beginning with those closest to the existing road network to minimise construction cost.
- ii) Engage telecom operators and the County Government of Embu to cross-check candidate zones against licensing, land tenure, and protected-area constraints, particularly near the Mount Kenya Forest Reserve boundary.
- iii) Extend the model with a gridded population-density layer (e.g. WorldPop) in a future iteration to quantify the number of residents each candidate site would newly serve.
- iv) Establish a periodic (e.g. annual) re-run of the Earth Engine script as OpenCellID tower data is updated, so that the suitability map remains current as new infrastructure is deployed.
- v) Consider extending the same reproducible workflow to neighbouring counties (Tharaka-Nithi, Kitui, Machakos) to support regional, rather than purely county-bounded, network-planning decisions.

10. References

- [1] OpenCellID. (2026). Open Database of Cell Towers, Kenya (MCC 639). <https://opencellid.org>
- [2] USGS Earth Explorer. (2024). Shuttle Radar Topography Mission (SRTM) 30 m Digital Elevation Model.
- [3] Humanitarian Data Exchange (HDX). (2024). Kenya Roads and Populated Places Datasets. <https://data.humdata.org>
- [4] GADM Database. (2024). Administrative Boundaries Kenya, Level 2 (Counties).
- [5] Google Earth Engine. (2026). USGS/SRTMGL1_003 DEM Collection and Terrain Algorithms Documentation.
- [6] Kenya National Bureau of Statistics (KNBS). (2019). Kenya Population and Housing Census, Vol. I Embu County.
- [7] QGIS Development Team. (2026). QGIS Geographic Information System. <https://qgis.org>

Appendices

Appendix A: Map Layout Portfolio

Figures 2–4 in Section 7 constitute the required minimum three-map portfolio: (A) Existing Telecom Infrastructure and Coverage Gaps, (B) Terrain Analysis and Slope Constraints, and (C) Optimal Cell Tower Site Suitability for 3G/4G/5G Network Expansion. All three maps share a common 1:456,269 scale, UTM Zone 37S (WGS 84) coordinate system, and consistent legend styling to allow direct visual comparison.

Appendix B: Earth Engine Script Structure

The Earth Engine script (task3_recovered) loads the county boundary, cell tower, population, and road assets; filters each to the county extent; clips and processes the SRTM DEM into elevation and slope layers; performs tower filtering and gap identification by radio type; and computes the final weighted suitability overlay. Intermediate and final rasters (EmbuElevationDEM, Embu_Slope, Embu 3G/4G/5G Suitability) were exported as Earth Engine tasks and ingested as Cloud Assets for use in QGIS.